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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT

credit card billing payment reminder system

Submitted by

Jai akash

25bcs119

Under the Guidance of

Mr.Aravind

TechnicalTrainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled "**credit card billing payment reminder system"**
submitted in partial fulfillment of the requirements for the course Java programming is a **record of my original work** carried out under the guidance of **Mr.Aravind**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Jaiakash

CERTIFICATE

This is to certify that the project entitled "credit card billing payment reminder system"is a bona fide work carried out by jaiakash -2025 Science, in partial fulfillment of the requirements for the course javaprogramming , BSC Computer science, **KGCollegeofArts & Science.**
This project has been completed under my supervision and guidance during the academic period

April2026.

Place: Coimbatore

Date:10\04\2026

Submitted for the Viva-Voce Examination held on 10/04/2026

Guide:
(Technical Trainer,KGCAS)
Mr.Aravind

HeadoftheDepartment
(Technical Head, KGCAS)
Mr.Boopal an Sundaramoorthy

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ABSTRACT

ACredit CardBilling Payment Reminder System serves as a sophisticated financial management tool designed to mitigate the risks associated with missed payment deadlines, such as exorbitant interest rates, late fees, and long-term damage to credit scores. At its core, the system functions by aggregating data from various credit card accounts to track billing cycles, minimum due amounts, and final payment dates. By leveraging automated triggers—ranging from push notifications and SMS alerts to email sequences— the system ensures that users are consistently informed of their financial obligations well in advance. This proactive approach allows individuals to plan their liquidity and cash flow more effectively, preventing the last-minute stress of managing multiple debt obligations across different banking platforms.

1.INTRODUCTION:

The Credit Card Billing Payment Reminder System is a software application designed to help users manage their credit card payments in an efficient and organized manner. In today's modern lifestyle, credit cards have become an essential part of financial transactions, including shopping, bill payments, and online purchases. However, with the increasing number of credit card users and multiple cards being used by a single individual, it becomes difficult to remember all the payment due dates. Missing these due dates can result in late payment fees, increased interest charges, and a negative impact on the user's credit score. This system is developed to address these challenges by providing timely reminders and maintaining proper records of billing details. Moreover, the system offers a centralized platform where users can store important information such as card details, billing cycles, due dates, and outstanding balances. It automatically tracks upcoming deadlines and sends notifications through various channels like SMS, email, or mobile app alerts. This reduces the need for manual tracking and minimizes the chances of human error. The system is designed to be user-friendly and reliable, ensuring that even non-technical users can easily operate it. In the future, it can be enhanced with advanced features such as automatic payment options, spending analysis, and personalized financial suggestions. Overall, the Credit Card Billing Payment Reminder System plays a crucial role in improving financial discipline, reducing financial stress, and helping users maintain
a good credit history.

1.1 Objective

The primary objective of the Credit Card Billing Payment Reminder System is to help users manage their credit card payments in a timely and efficient manner. It aims to provide automated reminders before the due date so that users do not miss their payments, thereby avoiding late fees, penalties, and high interest charges. The system is designed to maintain accurate records of billing information such as due dates, outstanding amounts, and payment history, which helps users stay organized and informed about their financial commitments. By doing so, it supports users in maintaining a good credit score and improving their overall financial stability. In addition, the system focuses on reducing the burden of manual tracking by automating the entire reminder process. It enhances user convenience by offering multiple notification methods such as SMS, email, and mobile app alerts, ensuring that users receive timely updates in a way that suits them best. The system also aims to handle multiple credit cards efficiently, making it suitable for users with several accounts. Furthermore, it can be extended with advanced features like automatic payments, spending analysis, and personalized financial suggestions. Overall, the objective of this system is to promote financial discipline, save time, and provide a reliable solution for effective credit card management.

1.2 Overview

The Credit Card Billing Payment Reminder System provides a comprehensive solution fo managing credit card payments in
a simple and efficient way. It is designed to help users keep track of multiple credit cards, thei billing cycles, due dates, and outstanding balances in one centralized platform. The system collect and stores all necessary information securely, allowing users to easily access and update thei details whenever required. By automating the reminder process, it ensures that users receive timel notifications before the due date, reducing the chances of missed payments. This makes the syster highly useful for individuals who struggle with remembering payment schedules or handlin multiple financial commitments. Moreover, the system enhances user experience by offerin flexible notification options such as SMS, email, and mobile app alerts. It minimizes manual effor and reduces the risk of errors in tracking payment deadlines. The design focuses on simplicity an usability, making it suitable for all types of users, including those with limited technica knowledge. In addition, the system can be expanded with advanced features like automatic paymer integration, spending analysis, and personalized financial advice. Overall, the system acts as reliable and efficient tool that supports better financial planning, improves credit management, an helps users maintain financial stability.

1.3 Features

TheUser Registration: Allows users to create and manage their accounts

securely. Multiple Card Management: Supports adding and managing

multiple credit cards.

Billin g Information Storage: Stores due dates, billing cycles, and

outstandingamount s.

Automa tic Reminders: Sends alerts before the payment due date.

Notification Sy stem: Provides reminders through SMS, email, or app

notifications.

Paymen t Tracking: Tracks paid and pending bills for better management.

User-Frien dly Interface: Easy to use for all types of users.

Data Sec urity: Ensures safe storage of user and card details.

Custom Rem inder Settings: Users can set reminder time and frequency

based on

2.PROGRAM ANALYSIS

The Credit Card Billing Payment Reminder System is designed to analyze user input and manage credit card payment information effectively. The program collects details such as user information, card number, billing date, due date, and outstanding amount, and stores them in a database. It processes this data to identify upcoming due dates and generates reminders accordingly. The system uses conditional logic to check whether a payment is pending, completed, or overdue, ensuring accurate tracking of each transaction. Furthermore, the program is structured into different modules such as data input, storage, processing, and notification. It ensures proper validation of user data to avoid errors and maintain reliability. The reminder scheduling function plays a key role by triggering alerts at the right time based on predefined settings. Overall, the program analysis shows that the system is efficient, user-friendly, and capable of handling multiple credit card accounts while ensuring timely notifications and better financial management. .

2.1Problem Definition

The Credit Card Billing Payment Reminder System is a software application that helps users track their credit card bills and sends reminders before the payment due date. It stores billing details, processes due date information, and generates alerts to ensure timely payments and avoid penalties.The Credit Card Billing Payment Reminder System is a software application developed to assist users in managing their credit card payments efficiently and accurately. It allows users to store important details such as card information, billing cycles, due dates, and outstanding amounts in a structured manner. The program continuously monitors these details and processes the data to identify upcoming payment deadlines. Based on this information, it automatically generates reminders and notifications to alert users before the due date, helping them avoid late payments, penalties, and additional interest charges. This system simplifies the task of handling multiple credit cards by providing all relevant information in one place. today's fast-paced world, many credit card users face difficulties in managing their payment schedules effectively. Due to busy lifestyles and the use of multiple credit cards, users often forget billing due dates, leading to late payments, penalties, and high interest charges. This not only causes financial loss but also negatively affects their credit score. Additionally, manual tracking of billing information is time-consuming and prone to errors. Therefore, there is a need for an automated system that can store billing details, monitor due dates, and send timely reminders to ensure users make payments on time and maintain proper financial discipline.

2.2 Project Overview

The Credit Card Billing Payment Reminder System is a project developed to simplify and automate the process of managing credit card payments. The main purpose of this project is to help users keep track of their billing details, due dates, and outstanding amounts in an organized manner. It provides a centralized platform where users can add and manage multiple credit card accounts easily. By automating reminders, the system ensures that users are notified before the payment deadline, reducing the risk of late fees and penalties. This project is designed with various modules such as user registration, card management, billing details, reminder scheduling, and notification system. Each module works together to provide accurate and timely information to the user. The system is user-friendly and reduces the need for manual tracking, making it efficient and reliable. Overall, the project helps improve financial discipline, saves time, and supports better management of credit card payments. The Credit Card Billing Payment Reminder System is a software- based project developed to provide an efficient and organized solution for managing credit card payments. In today's fast- paced lifestyle, many individuals use multiple credit cards for various transactions, making it difficult to remember all payment due dates. This project aims to overcome such challenges by offering a centralized platform where users can store and manage all their credit card details, including billing cycles, due dates, and outstanding balances. By maintaining this information in a structured database, the system ensures easy access, better organization, and improved financial tracking for users.

2.3 Scopes and Limitations:

The scope of the Credit Card Billing Payment Reminder System is to provide a simple and effective solution for managing credit card payment schedules. It helps users track due dates and receive timely reminders, ensuring that payments are made without delay. The system is designed to be user-friendly and easy to operate, making it suitable for individuals with basic technical knowledge. It demonstrates how programming can be used to solve real-life financial problems and can be applied as a foundation for more advanced systems. Additionally, the system can be expanded in the future by integrating features such as automatic notifications, database management, and mobile application support, which would further enhance its usefulness and efficiency. However, the system also has certain limitations. It relies on manual input of due dates and current dates, which may lead to errors if incorrect data is entered. The current version does not support automatic reminders through SMS or email, limiting its real-time functionality. It also lacks a database to store user information and payment history, meaning that data is not saved permanently. Furthermore, the system is basic in design and does not include advanced features like online payment integration or detailed financial analysis. Despite these limitations, it serves as a simple and effective prototype for a credit card reminder system

Despite its limitations, the system provides a strong base for future improvements and real-world applications. With further development, it can be transformed into a fully automated and intelligent system that integrates with banking services and mobile platforms. Enhancements such as real-time notifications, cloud storage, and advanced security features can significantly improve its performance and usability. This makes the project highly adaptable and valuable for modern financial management needs. Furthermore, the system can be extended to support additional functionalities such as multi-language support and user-friendly graphical interfaces to improve accessibility. It can also include alert prioritization and reminder repetition features to ensure that important payments are not missed. With continuous improvements and updates, the system can become more robust, scalable, and suitable for a wide range of users in different environments.

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actua implementation. In this phase, the system is divided into different modules, and the workflow of the application is clearl defined. Proper system design helps in organizing the program efficiently and makes the development process easier an more systematic. The Savings Goal Planning and Financial Progress Tracking System is designed as a menu- drive application where users can set savings goals, record income and expenses, and track their financial progres automatically. The system processes user inputs and provides accurate calculations along with a clear summary o financial data. The system is divided into several modules to perform specific tasks. Each

3.1 Modules

The main modules of the **creditcardbillingpayment reminder system 1.Menu Display Module** are explained below:
Example Goal:

1.Buy Mobile ₹20,000 (5 months)
2.Laptop Purchase ₹50,000 (10 months)
3.Trip Planning ₹15,000 (3 months)
4.Education Fund ₹30,000 (8 months)
5.Emergency Savings ₹10,000 (4 months)

Thegoal details aredisplayed whenthe program starts so that the user can easily view and understand their financial targets. This module forms the base for tracking savings and progress in the system. This module is responsible for displaying and managing the user's savings goals along with their target amount and time period. It helps the user clearly define what they want to achieve financially. The system allows the user to enter details such as the goal name, target amount, and duration. This information is stored and used throughout the program to track progress.

2.Income and Expense Entry Module

This module allows the user to select food items from the menu and enter the quantity ordered by the customer. Functions of this module include: • Accepting user input for item selection• Accepting quantity of selected items• Storing order details temporarily The system allows users to select multiple food items before generating the final bill.2. Order Selection Module This module allows the user to select food items from the menu and enter the quantity ordered by the customer.

3.Bill Calculation Module

The Billing Module is responsible for managing all credit card billing information within the system. It stores details such as billing date, due date, total amount due, and transaction summary for each credit card. This module helps organize and maintain accurate records of user payments.processes the billing data to ensure that the correct amount and due date are displayed to the user. The module also supports tracking of outstandingbalances and updates payment status after transactions are completed. Overall, the billing module plays a key role in ensuring accurate billing information and supports timely payment management.

4. Invoice Generation Module

The Invoice Generation Module is responsible for creating and displaying a detailed bill for the user based on credit card usage. It collects information such as transaction details, total amount due, billing date, and due date. The module processes this data to generate a clear and organized invoice, which can be viewed or saved by the user.
This module ensures accuracy in billing by calculating the total payable amount and including any additional charges if applicable. It also helps users understand their expenses by presenting the information in a structured format. Overall, the invoice generation module improves transparency and makes it easier for users to review and manage their credit card payments.

5. Exit Module

The Exit Module is responsible for safely closing the Credit Card Billing Payment Reminder System. It allows the user to exit the app lication after completing all required operations. Before termination, the system ensures that all processes are completed and any necessary data is properly handled.

This module improves user experience by providing a clear option to end the program without errors or data loss. It ensures smooth termination of the system and prevents unexpected crashes. Overall, the exit module plays an important role in maintaining the stability and reliability of the application.

4.METHODOLOGY

The development of the Credit Card Billing Payment Reminder System follows a systematic approach to ensure proper design and functionality. First, the problem is identified, focusing on the difficulty users face in remembering payment due dates. Then, a simple algorithm is designed to compare the due date with the current date. Based on this algorithm, the system is developed using a programming language like Java, where user inputs are collected and processed using conditional statements. Next, the system is implemented by dividing it into modules such as input handling, data processing, and output generation. After implementation, testing is carried out using different input values to verify accuracy and reliability. Finally, the system is evaluated and improved based on test results. This methodology ensures that the system is efficient, user-friendly, and capable of providing correct reminders for credit card payments.

4.1.Problem Analysis

TheCredit Card BillingPayment Reminder System is designed to analyze user-provided data and generate appropriate reminders based on payment status. The program takes input values such as the due date and current date from the user and processes them using conditional statements. By comparing these values, the system determines whether the payment is upcoming, due today, or overdue, and displays the corresponding message. The program follows a simple structure consisting of input, processing, and output stages. In the input stage, user data is collected using the Scanner class. In the processing stage, logical conditions (if-else statements) are applied to evaluate the relationship between the two dates. Finally, in the output stage, the result is displayed to the user in the form of reminder messages. This analysis shows that the program is efficient, easy to understand, and effectively solves the problem of tracking credit card payment deadlines.

4.2 System Design

1.Simple and user-friendly system structure.
2.Input module to collect due date and current date.
3.Processing module to compare dates using conditions.
4.Output module to display reminder messages.
5.Modular design for easy understanding and development.

4.3.Program Development

The Credit Card Billing Payment Reminder System is developed using a simple and structured programming approach. The development process begins with identifying the problem and designing a basic algorithm to compare the due date and curren date. The program is implemented using a programming language like Java, where user input is taken through the scanner class, and conditional statements are used to determine the payment status. During development, the program is divided into steps such as input collection, processing, and output generation. The logic checks whether the current date is less than, equa to, or greater than the due date and displays the appropriate reminder message. The program is tested with different input values to ensure accuracy and reliability. Overall, the development process focuses on simplicity, clarity, and correct functionality, making the system easy to understand and use. The development of the Credit Card Billing Payment Reminder System follows a step-by-step approach to ensure proper functionality and ease of use. Initially, the problem is analyzed, and simple algorithm is designed to compare the due date with the current date. Based on this logic, the program is coded using Java, where user inputs are taken through the Scanner class. Conditional statements (if-else) are used to evaluate the dates an generate appropriate reminder messages. The program is structured into key stages

4.4.Testing and Validation

The Credit Card Billing Payment Reminder System is tested using different input values to ensure correct functionality. Various test cases are applied, such as when the current date is less than, equal to, and greater than the due date. The system correctly displays messages like “Payment due soon,” “Payment due today,” and “Payment overdue,” based on the conditions. This confirms that the conditional logic in the program works accurately. Validation is performed by checking whether the user inputs are accepted correctly and the output is generated as expected. The system ensures that only valid numerical inputs are processed, reducing errors during execution. Overall, the testing and validation process proves that the system is reliable, accurate, and performs efficiently under different scenarios.

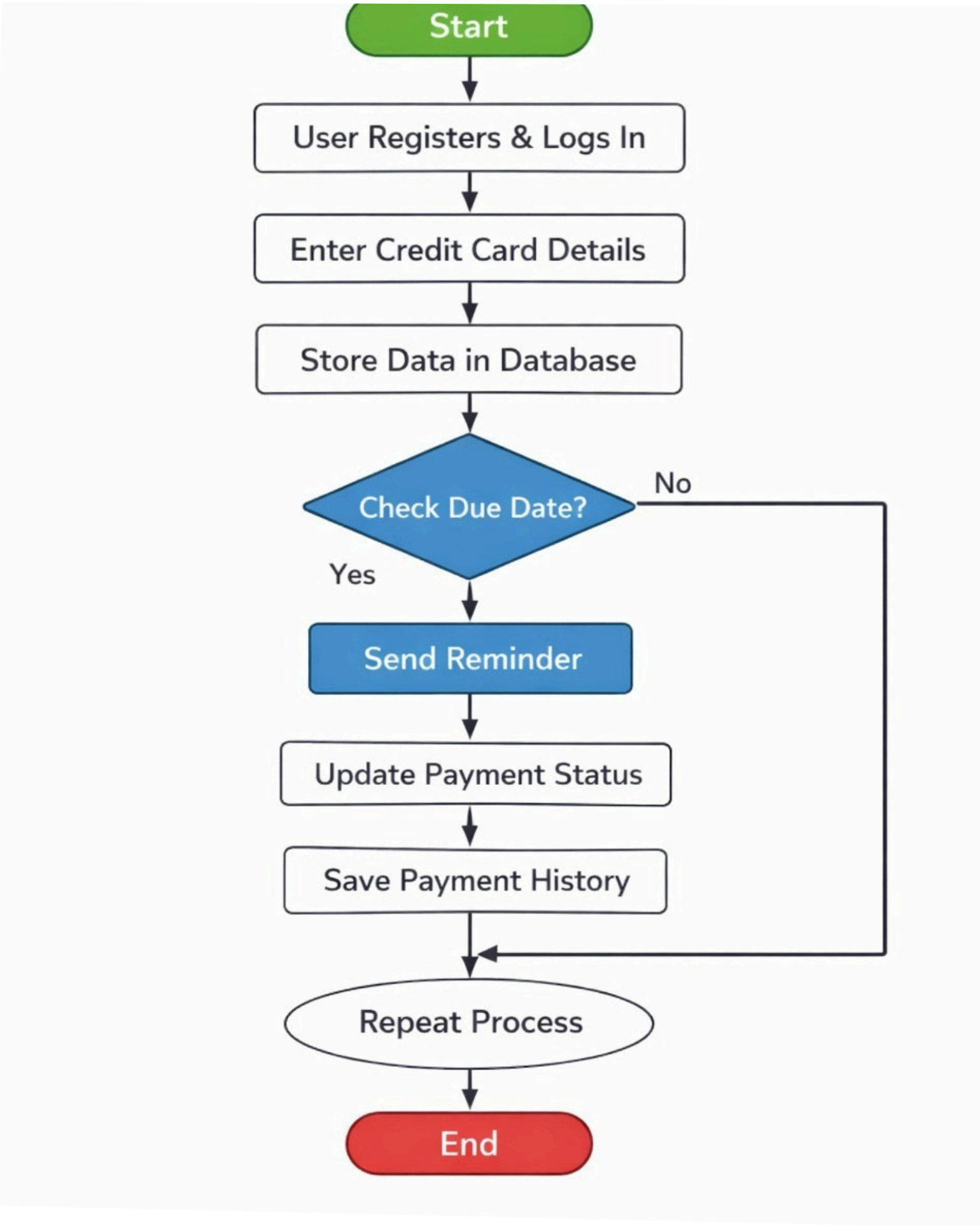
4.5. Result and Documentation

The Credit Card Billing Payment Reminder System produces accurate results by sending timely reminders and tracking payment status effectively. It helps users avoid late payments and manage their credit card bills efficiently. All user details, billing information, and payment records are stored properly for future reference. The documentation clearly explains the system design, modules, algorithm, and overall working process, making it easy to understand, implement, and maintain the system.

4.6 Alogorithm

1.Start the process.
2.User registers and logs into the system.
3.Enter credit card details.
4.Store the entered data in the database.
5.check the payment due date.
6. Ifhe due date is reached: Send reminder to the user. Update payment status. Save payment history Else (if due date not reached):
7.skip reminder and continue process.
8.Repeat the process for next cycle. 9.End the process.

4.7Flowchart



5.SOURCE CODE;

```
import java.util.Scanner;

public class Reminder {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int due, current;

        System.out.print("Enter due date: ");
        due = sc.nextInt();

        System.out.print("Enter current date: ");
        current = sc.nextInt();

        if (current >= due - 3 && current <= due)
            System.out.println("Reminder: Pay your bill!");
        else if (current > due)
            System.out.println("Overdue!");
        else
            System.out.println("No reminder.");

        sc.close();
    }
}
}
```

```
else if (currentDate == dueDate) {
            System.out.println("Reminder: Payment due
today!");
        }
        else {
            System.out.println("Alert: Payment overdue!");
        }


        sc.close();
    }
}
```

INPUT &OUTPUT SCREEN

```
Reminder

Enter due date: 20
Enter current date: 18

Reminder: Pay your bill!
```

```
Reminder

Enter due date: 20
Enter current date: 20

Reminder: Pay your bill!
```

```
Reminder

Enter due date: 20
Enter current date: 25

Overdue!
```

```
Reminder

Enter due date: 20
Enter current date: 10

No reminder.
```

1. DISCUSSION & CONCLUSION

7.1 Discussion

credit card billing payment reminder system is a critical financial tool designed to bridge the gap between a bank's receivables and a customer's busy schedule. By leveraging automated omni-channel alerts—such as SMS, push notifications, and emails—these systems ensure that users are prompted at strategic intervals before their due date. For the consumer, this automation is essential for avoiding late payment fees, high-interest charges, and potential damage to their credit score. For financial institutions, it serves as a proactive risk management strategy that maintains steady cash flow and reduces the operational costs associated with manual debt collection. Ultimately, an effective reminder system fosters financial discipline and strengthens the trust between the lender and the cardholder..

7.2 Conclusion

In conclusion, a credit card billing payment reminder system is an indispensable tool in modern financial management. By integrating automated alerts across multiple digital channels, it transforms a potentially stressful manual task into a seamless, user-centric experience. For consumers, these systems provide a critical safety net that protects their credit health and financial standing; for lenders, they serve as a highly efficient mechanism to ensure timely collections and operational stability. As financial technology continues to evolve, the shift toward more personalized and AI-driven reminder systems will likely further reduce delinquency rates and promote long-term fiscal responsibility.

8 . REFERENCES

The following resources were used during the development of this project:

Book:

Let java : ***beginner-friendly book to learn Java programming concepts used in your project.***

websites:

Learn HTML, CSS, JavaScript step by step for building your website. Provides coding examples, Java programs, and project ideas.